Attention Is All You Need

(Attention instead of recurrence)

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Key points of 'Attention Is All You Need' paper



Problem Statements

RNN and LSTM model limitations:

1. **Sequential processing:** process one input token at a time – no parallel processing – makes training slow.
2. **Struggle with long term dependencies**: by the time network reach the end of sequence, much of early information was lost.



Approach

1. **Transformer architecture**: add *attention* layer which let all tokens directly share information each other. (Attention acts like a communication layer inside the network allow faster-parallel processing and overcome long term dependencies).



Outcomes

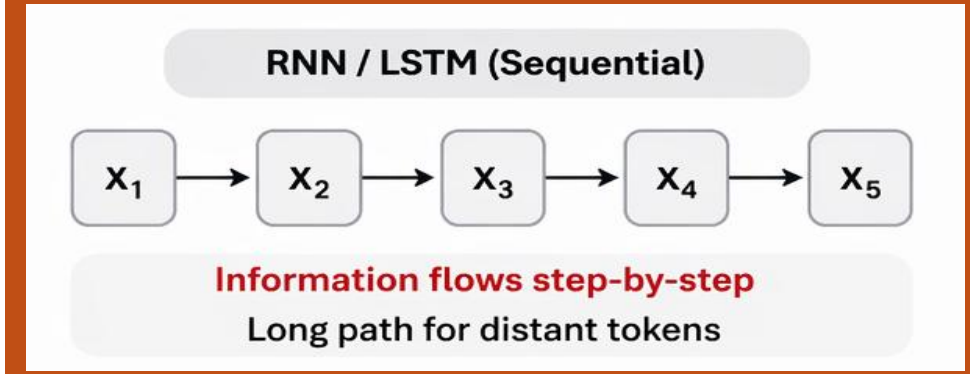
1. Achieves SOTA performance on **WMT14** English-German & English-French machine translation benchmark datasets.
2. **Attention** alone is enough to sequential modeling; efficient training cost

* ***Token*** is a piece of text given to the model as input. It can be a word, part of a word, or punctuation mark. A sentence is made of many tokens.

Why the Transformer model proposed? Its advantages

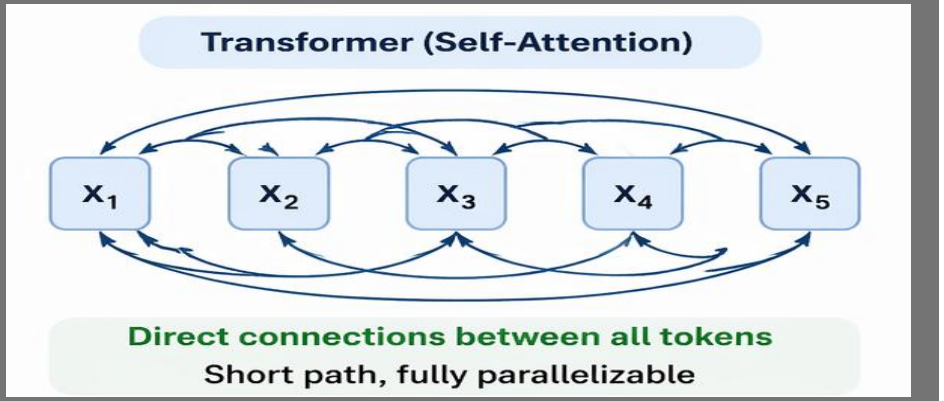
Recurrence and convolutional based earlier RNN and LSTM model struggle with efficiency and long-range dependency.

RNN & LSTM model processing of input tokens



- Only near two token can share information at the same time
- Longer training time
- Information loss for long term dependencies
- Lower accuracy

Transformer model processing of input tokens



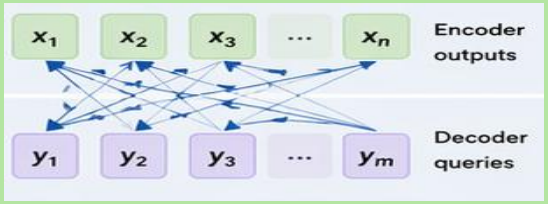
- All input tokens share information at the same time
- Training time significantly reduced
- No information loss for long term dependencies
- Higher accuracy than previous SOTA model

Transformer architecture (One attention mechanism, three roles)

- ***The entire Transformer architecture developed based on attention layers.***
- ***The same multi-head attention mechanism is reused in three roles:***
  Encoder self-attention, decoder self-attention, encoder-decoder attention

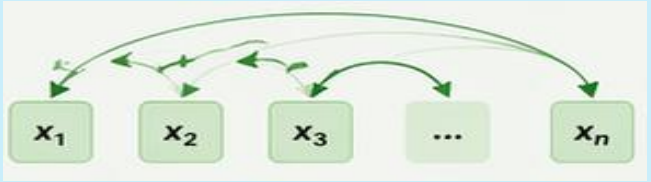
Encoder-Decoder attention

- Decoder queries attend to encoder outputs (keys/values)
- Aligns the input with the output.



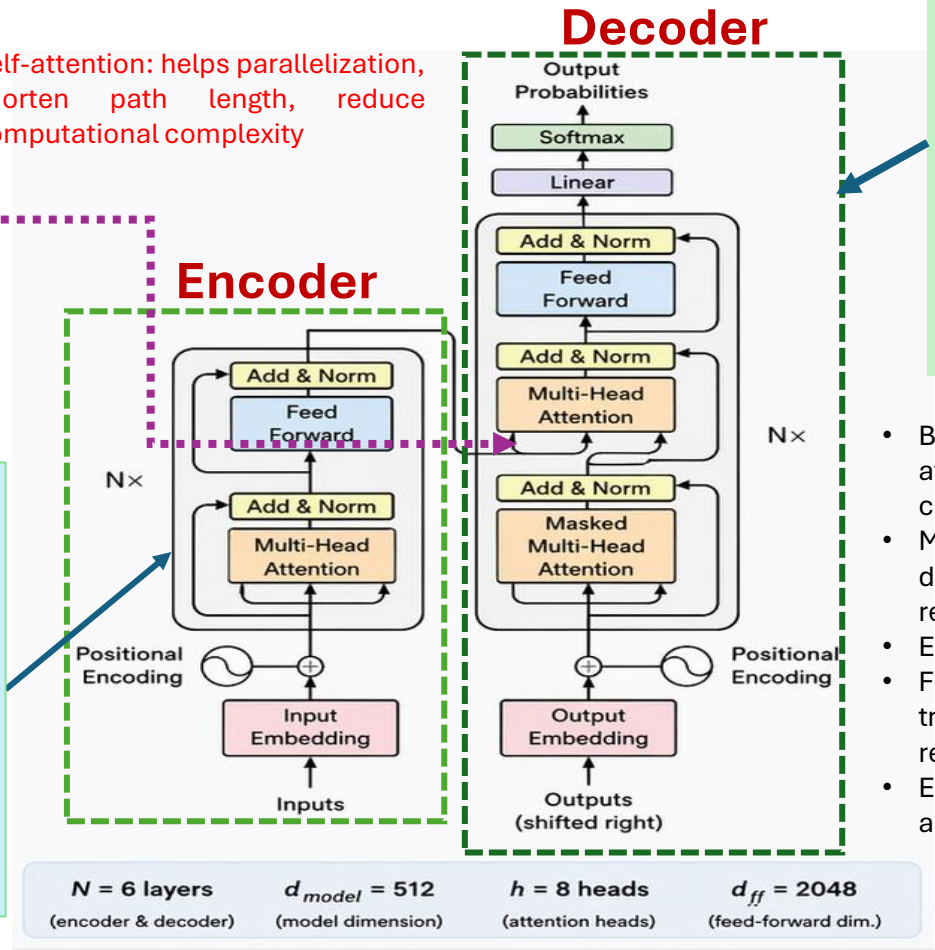
Encoder self-Attention

- Every source token interact to all source tokens.
- Captures contextual relationships of the tokens.



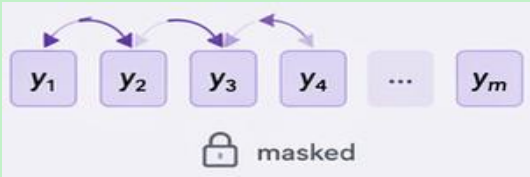
Tokens x1, x2, x3 ..... xn directly interact each other

Self-attention: helps parallelization, shorten path length, reduce computational complexity



Decoder self-attention

- Every target token interact only to earlier tokens and itself
- Future tokens are masked

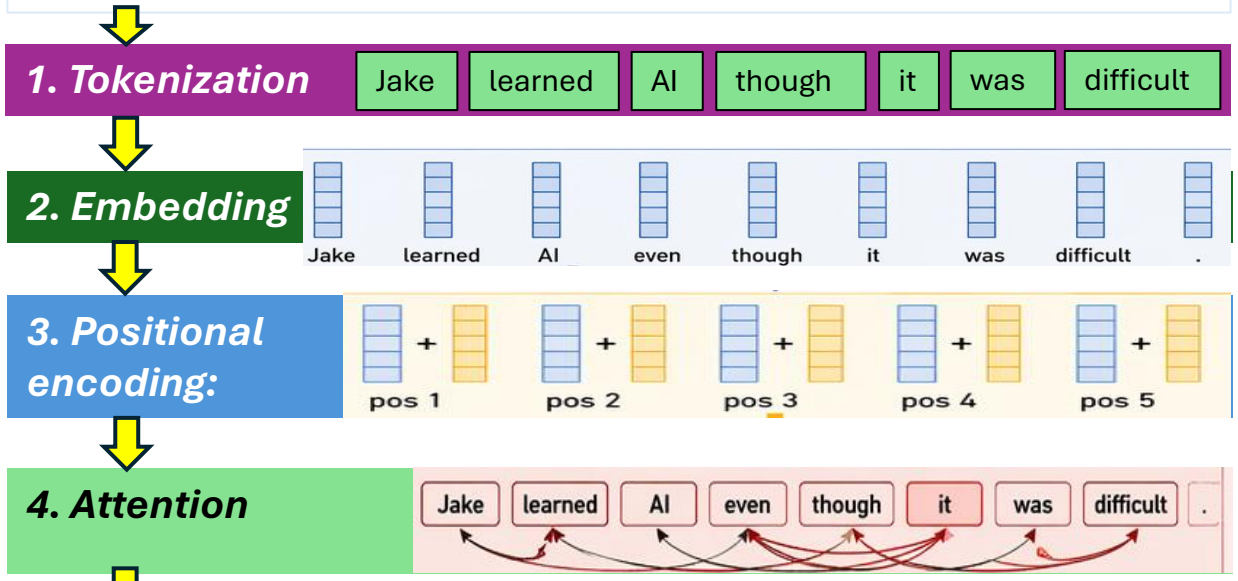


- y2 can only interact with y1 and itself, not with y3

- Both encoder & decoder network used stacked self-attention (N=6), feed forward block, residual connections and layer normalization.
- Multi-heads (h=8) attention lets the model learn different linguistic relations; syntax, phase structure, reference, long distance dependencies.
- Each head dimension is 512/8 = 64
- Feed forward network perform non-linear transformation at each position, that helps each token representation more expressive non-linear processing.
- Embedding & softmax: keeps architecture efficient & align input/output representation spaces.

Encoder and Decoder: How they work

Example input sentence: Jake learned AI even though it was difficult

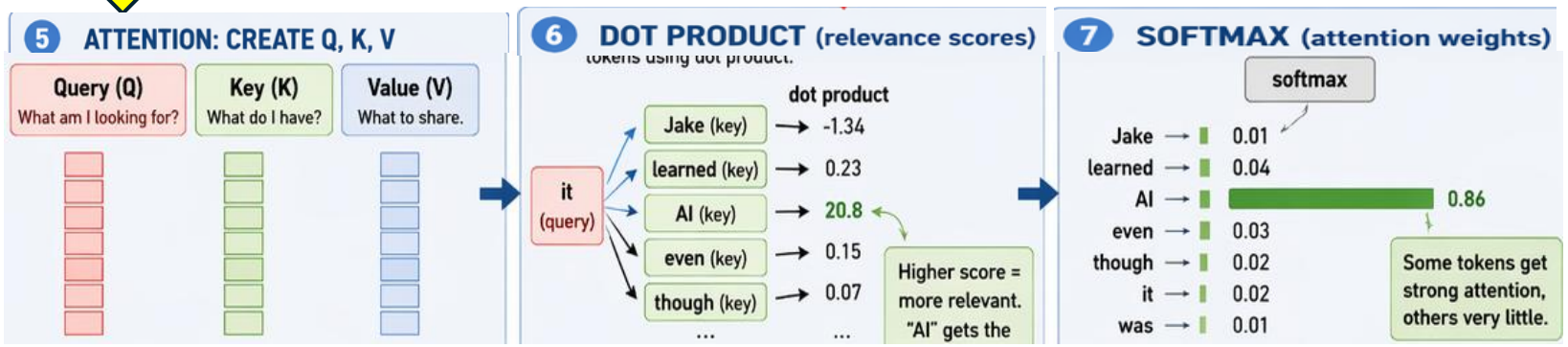


1. Sentence split into tokens

2. Tokens convert into numerical embeddings

3. N layer process embeddings repeatedly
Add position information to understand order

4. All tokens exchange information parallelly

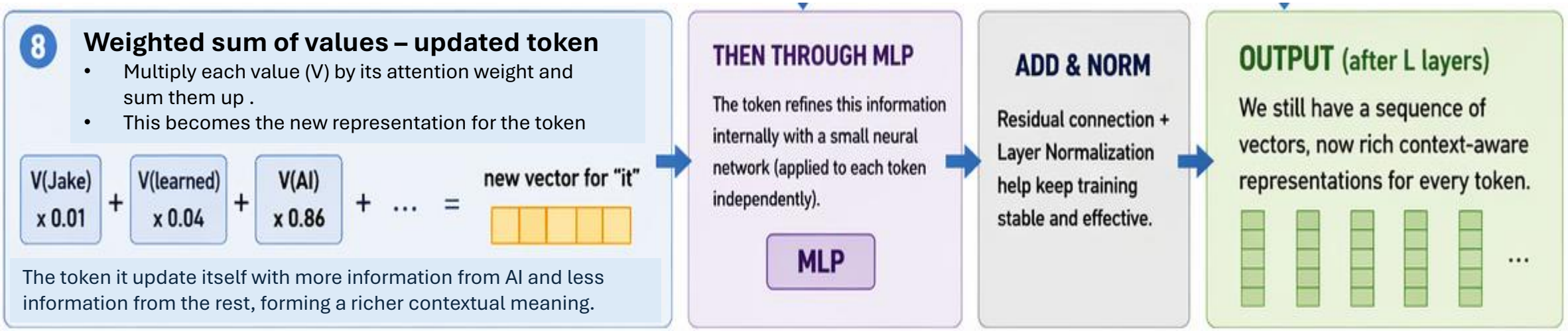


5. Generate Query (Q), Key (K) and value (V) vectors for each token

6. Compare query and key with all other tokens using dot product to decide which tokens are relevant.

7. Normalize the score with softmax function to turn them into attention weight.

Encoder and Decoder: How they work (continue....)



- This process gives us a new context-aware representation for each token.
- One that blends the most relevant information from the rest of the sequence.

Mathematically the paper express the process in matrix form:

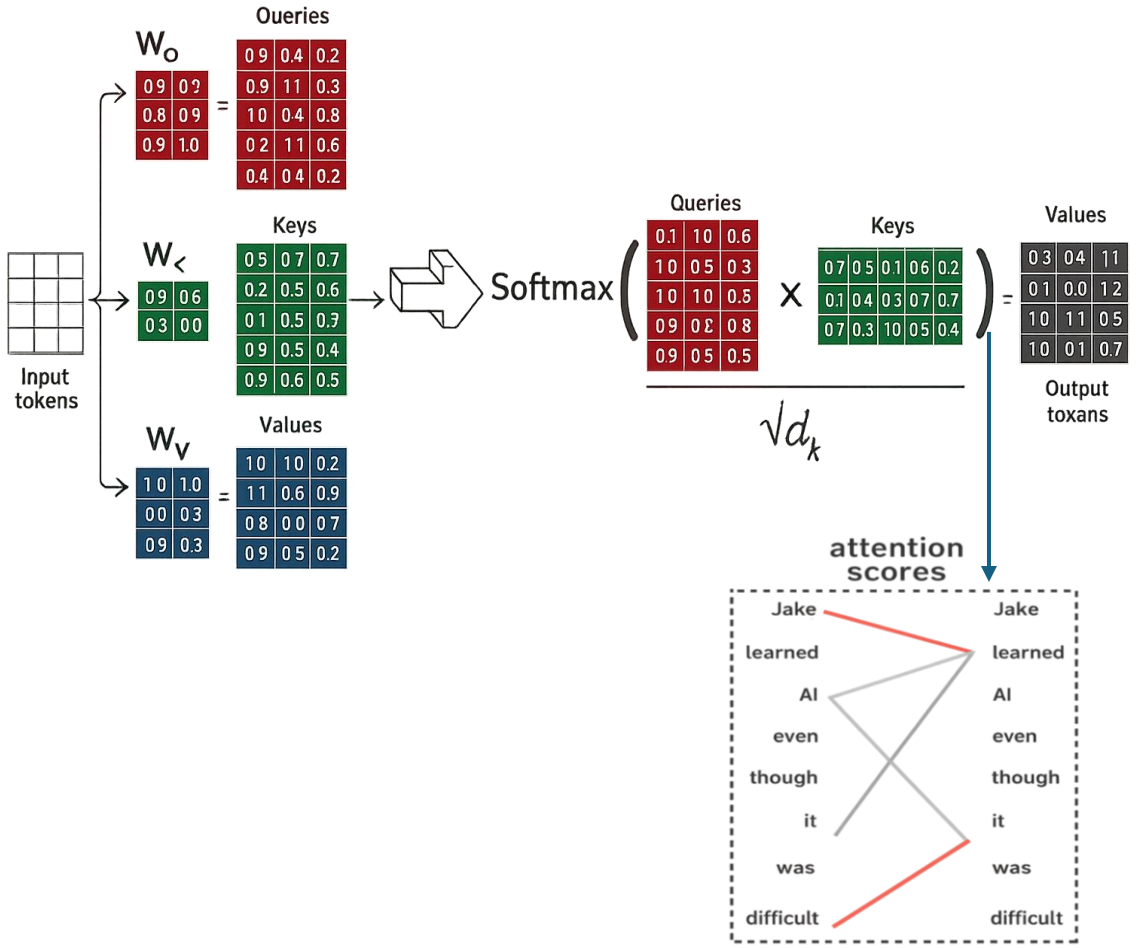
$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_{lc}}}\right) V$$

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Encoder and Decoder: How they work (continue...)

Quick overview of the model working process with bullet points

- Instead of looping through tokens one by one, the model stacks all queries, keys and values into matrices and performs the dot product and weighted sums simultaneously.
- This means every token communicates with every other token in a single set of parallel matrix operations, which is efficient and fully differentiable.
- At the very beginning of training, all the parameters are random, therefore all the representations are meaningless.
- The model has no idea what to look for or what to offer, but as training progresses, the parameters that produces the queries, keys and values are optimized.
- Overtime, the attention layer learns meaningful patterns. For instance, vers like “learned” start query their subjects and pronouns like “it” learn to look forward relevant nouns like AI.
- Other details masked, multi-head and cross-attention just modify how attention is calculated.
- This variations help the model handle sequence order, enforce causality and combine information from different sources.
- So, the transformer is a powerful way of stacking neural layers that allows dynamic communication between sequence element.
- This design turns out to be incredibly general. It can be adopted to support different tasks like translation, summarization and text generation.
- Moreover, it can also extends beyond language to images, audio and even code.
- Whenever data can be viewed as a sequence of elements that need to interact Transformer shine.
- Transformer can be used in encoder-decoder setups like in the original paper for translation, or decoder only models like GPT for text generation.
- Remember, a Transformer is a network that lets its inputs talk to each other. Its not magic, its communication. That’s why attention really is all we need.



Training data and hardware setup

Training Data

Dataset	# Sentence Pairs	Vocabulary
WMT14 English–German	~4.5 million	Shared BPE ~37,000 tokens
WMT14 English–French	~36 million	Word-piece 32,000 tokens

Standard MT benchmarks; large-scale parallel data.

Hardware and Schedule

Model	Sec/Step	Total Steps	Time
Base Model	~0.4	100,000	~12 hours
Big Model	~1.0	300,000	~3.5 days

Trained on 1 machine with 8 NVIDIA P100 GPUs.

Optimizer

Optimizer	Learning Rate Schedule	Note
Adam (β_1=0.9, β_2=0.98, ε=10^{-9})	Inverse square root with warmup	Stable updates

Regularization

Technique	Value	Note
Label Smoothing	0.1	Prevent overconfidence
Dropout	0.1	Reduce overfitting

Adam + warmup for stable training; regularization improves generalization.

Result

Table: Transformer achieves higher BELU score than previous SOTA model on EN-DE & EN-FR WMT14 benchmark datasets at a very lower training cost.

Model	BLEU		Training Cost (FLOPs)	
	EN-DE	EN-FR	EN-DE	EN-FR
ByteNet [18]	23.75			
Deep-Att + PosUnk [39]		39.2		$1.0 \cdot 10^{20}$
GNMT + RL [38]	24.6	39.92	$2.3 \cdot 10^{19}$	$1.4 \cdot 10^{20}$
ConvS2S [9]	25.16	40.46	$9.6 \cdot 10^{18}$	$1.5 \cdot 10^{20}$
MoE [32]	26.03	40.56	$2.0 \cdot 10^{19}$	$1.2 \cdot 10^{20}$
Deep-Att + PosUnk Ensemble [39]		40.4		$8.0 \cdot 10^{20}$
GNMT + RL Ensemble [38]	26.30	41.16	$1.8 \cdot 10^{20}$	$1.1 \cdot 10^{21}$
ConvS2S Ensemble [9]	26.36	**41.29**	$7.7 \cdot 10^{19}$	$1.2 \cdot 10^{21}$
Transformer (base model)	27.3	38.1	**$3.3 \cdot 10^{18}$**	
Transformer (big)	**28.4**	**41.8**	$2.3 \cdot 10^{19}$	

Table: Transformer achieves higher BELU and lower training cost compared to previous best SOTA method.

Model	En–De (BLEU)	En–Fr (BLEU)	Training Cost
ConvS2S Ensemble (Previous SOTA)	26.36	41.29	7.7×10^{19} (EN-DE) 1.2×10^{21} (EN-FR)
Transformer (base)	27.3	38.1	3.3×10^{18} (Vey low)
Transformer (big)	**28.4 (+2.04)**	**41.8 (+0.51)**	2.3×10^{19} (low)

- The Transformer (big) achieved 28.4 BLEU on WMT14 English–German and 41.8 BLEU on WMT14 English–French, improving over the previous state-of-the-art ConvS2S ensemble by +2.04 BLEU and +0.51 BLEU, respectively.
- It also required lower training cost: on English–German, the Transformer used about 2.3 × 10^19 FLOPs versus 7.7 × 10^19 for the previous best model, and on English–French, it used about 2.3 × 10^19 FLOPs versus 1.2 × 10^21, showing much higher computational efficiency.
- This showed that the Transformer is a highly general model for translation, summarization, and text generation, and it also extends beyond language to images, audio, and code.

***BLEU (Bilingual Evaluation Understudy) evaluate machine translation quality; how similar the model's output is to a reference (human) translation.**

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Limitations

The original Transformer was a breakthrough, but the paper also reveals constraints that motivated later research directions.

Key limitations

Expensive training setup
The paper relies on large WMT datasets and 8 NVIDIA P100 GPUs, which limits accessibility.

Order is not modeled inherently
Without recurrence or convolution, sequence order must be injected through positional encoding.

Evaluated mainly on translation
The original validation focuses on machine translation benchmarks, leaving broader tasks open.

Question & Answer



Thank You!